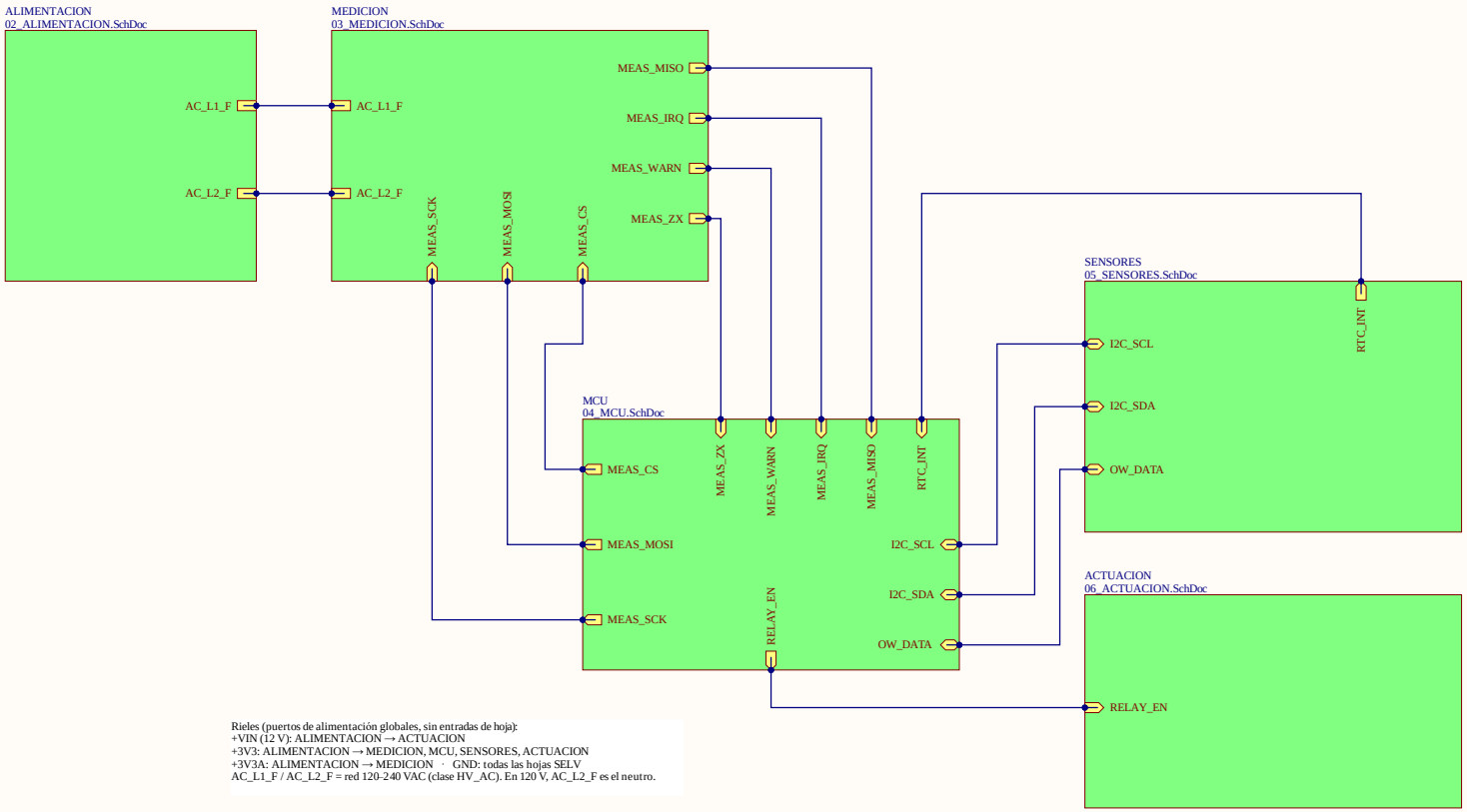
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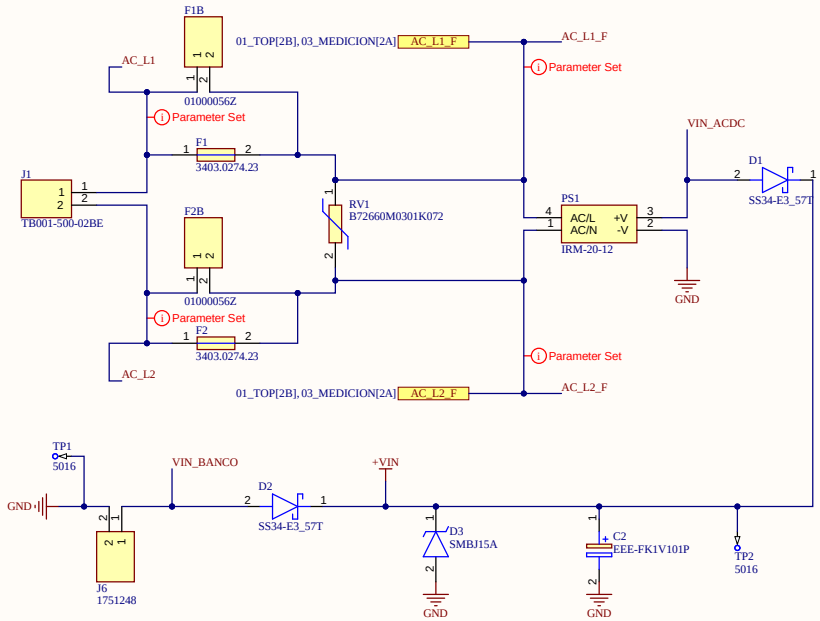
Rieles (puertos de alimentación globales, sin entradas de hoja):
+VIN (12 V): ALIMENTACION → ACTUACION
+3V3: ALIMENTACION → MEDICION, MCU, SENSORES, ACTUACION
+3V3A: ALIMENTACION → MEDICION · GND: todas las hojas SELV
AC_L1_F / AC_L2_F = red 120-240 VAC (clase HV_AC). En 120 V, AC_L2_F es el neutro.

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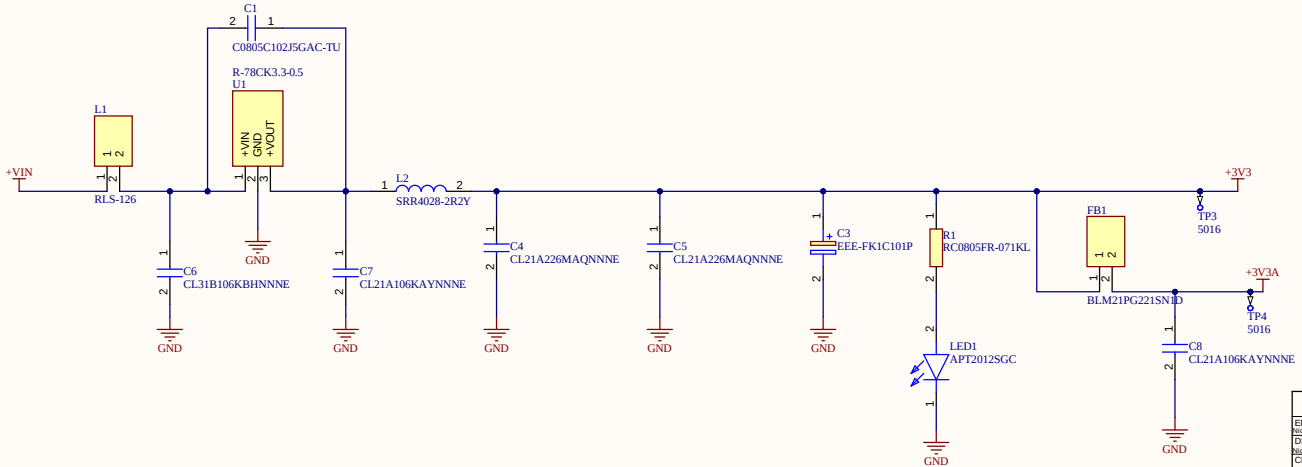
PROJECT: Control de Carga
PROJECT REVISION: 0.1
DOCUMENT REVISION: Not in version control
DESIGN BY: Nicolás Gómez Ramírez
TITLE: *
SIZE: A3
CAGE CODE:
DWG NO.:
REV:
SCALE:
FILE NAME: 01_TOP.SchDoc
SHEET 1 OF 6

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Alimentar desde breaker exclusivo de 2-6 A (bipolar en 240 V). J1: 1 = L1/L, 2 = L2/N. J6: 12 V de banco (9-15 V).

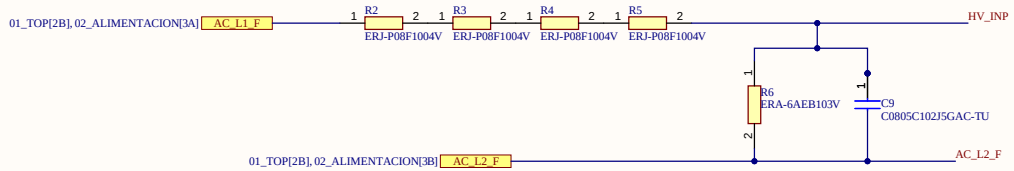


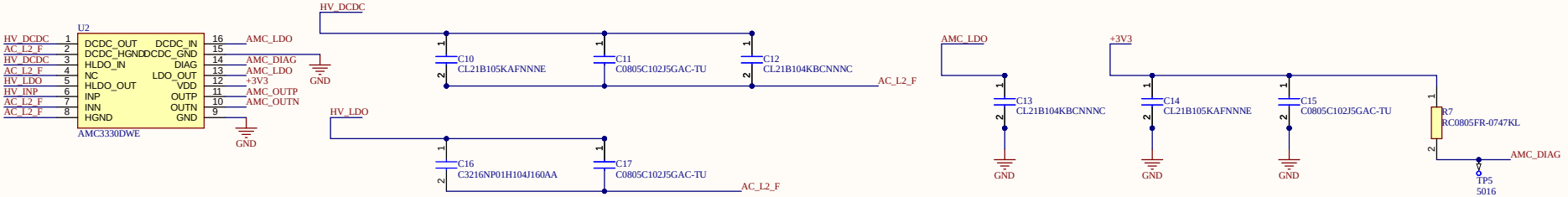
APPROVALS	DATE
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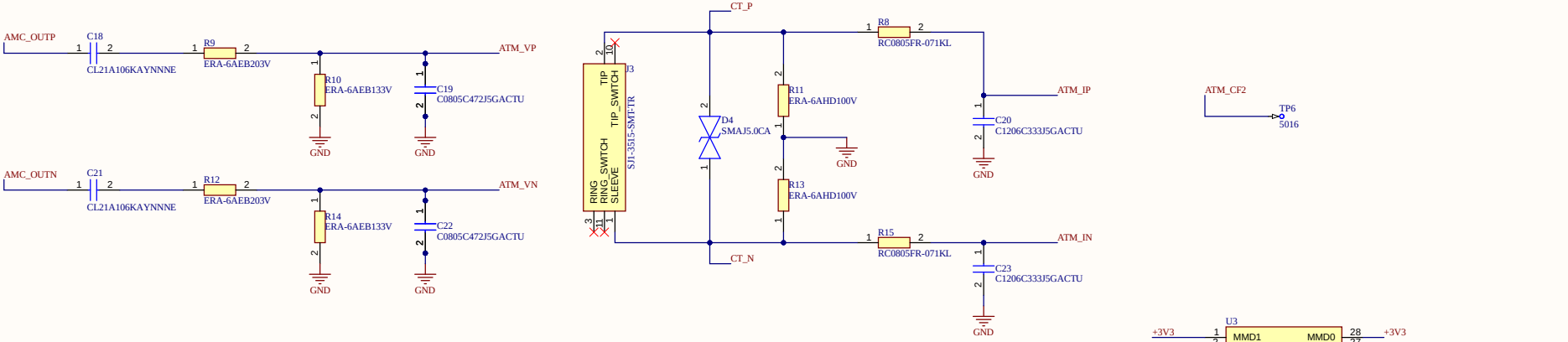
PROJECT: Control de Carga
PROJECT REVISION: 0.1
DOCUMENT REVISION: Not in version control
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TITLE: *
SIZE: A3
CAGE CODE:
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FILE NAME: 02_ALIMENTACION.SchDoc
SHEET 2 OF 6

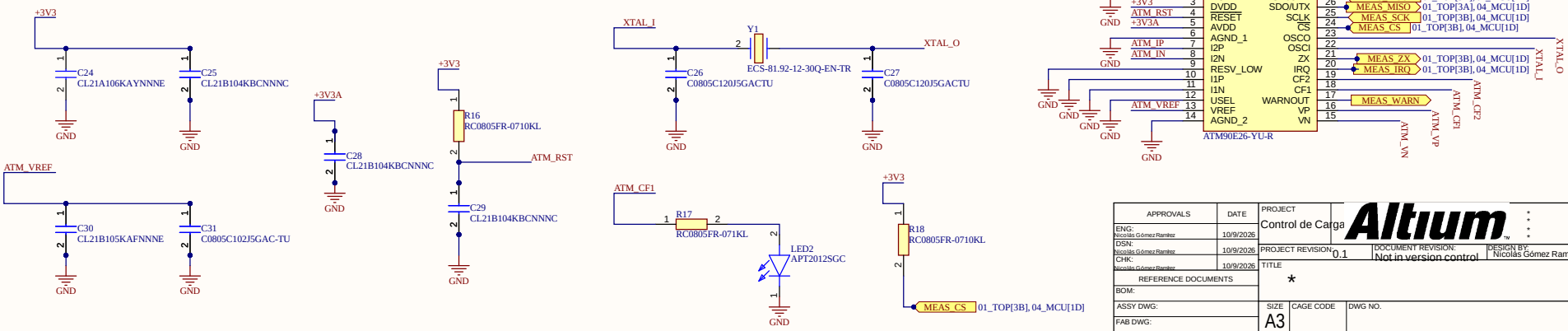
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CHK: Nicolás Gómez Ramírez	10/9/2026	TITLE *	DESIGN BY: Nicolás Gómez Ramírez
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PCB DWG:		SCALE:	FILE NAME 03_MEDICION.SchDoc SHEET 3 OF 6

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TOP[4B], 03_MEDICION[5C] MEAS_MISO
TOP[4B], 03_MEDICION[5D] MEAS_IRQ
TOP[3C], 03_MEDICION[5C] MEAS_SCK
TOP[3C], 03_MEDICION[5C] MEAS_MOSI
ION[4D], 03_MEDICION[5C] MEAS_CS
P[4C], 06_ACTUACION[1D] RELAY_EN
TOP[4B], 03_MEDICION[5D] MEAS_WARN
TOP[3B], 03_MEDICION[5C] MEAS_ZX
TOP[4B], 05_SENSORES[2D] RTC_INT

01_TOP[4C], 05_SENSORES[1D] I2C_SDA
01_TOP[4C], 05_SENSORES[1D] I2C_SCL
01_TOP[4C], 05_SENSORES[1D] OW_DATA

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PROJECT: Control de Carga
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DOCUMENT REVISION: Not in version control
DESIGN BY: Nicolás Gómez Ramírez
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SHEET 4 OF 6

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01_TOP[4B], 04_MCU[2D] I2C_SCL

01_TOP[4B], 04_MCU[2D] I2C_SDA

01_TOP[4B], 04_MCU[2D] OW_DATA

01_TOP[5B], 04_MCU[1D] RTC_INT

APPROVALS	DATE
ENG: Nicolás Gómez Ramírez	10/9/2026
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PROJECT: Control de Carga

PROJECT REVISION: 0.1

DOCUMENT REVISION: Not in version control

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SHEET 5 OF 6

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01_TOP[4C], 04_MCU[1D] RELAY_EN

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PROJECT: Control de Carga

PROJECT REVISION: 0.1

DOCUMENT REVISION: Not in version control

DESIGN BY: Nicolás Gómez Ramírez

TITLE: *

SIZE	CAGE CODE	DWG NO.	REV
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SCALE:

FILE NAME: 06_ACTUACION.SchDoc

SHEET 6 OF 6

Comment	Description	Designator	Footprint	LibRef	Quantity
C0805C102J5GAC-TU	Capacitor	C1, C9, C11, C15, C17, C31	CAPC2012X88N	C0805C102J5GAC-TU	6
EEE-FK1V101P	Capacitor Polarised	C2	EEHZT1E331UP	EEE-FK1V101P	1
EEE-FK1C101P	Capacitor Polarised	C3	EEE1AA101SP	EEE-FK1C101P	1
CL21A226MAQNNNE	Capacitor	C4, C5	CAPC2012X140N	CL21A226MAQNNNE	2
CL31B106KBHNNNE	Capacitor	C6	CAPC3216X180N	CL31B106KBHNNNE	1
CL21A106KAYNNNE	Capacitor	C7, C8, C18, C21, C24	CAPC2012X145N	CL21A106KAYNNNE	5
CL21B105KAFNNNE	Capacitor	C10, C14, C30	CAPC2012X88N	CL21B105KAFNNNE	3
CL21B104KBCNNNC	Capacitor	C12, C13, C25, C28, C29	CAPC2012X88N	CL21B104KBCNNNC	5
C3216NP01H104J160AA	Capacitor	C16	CAPC3216X180N	C3216NP01H104J160AA	1
C0805C472J5GACTU	Capacitor	C19, C22	CAPC2012X88N	C0805C472J5GACTU	2
C1206C333J5GACTU	Capacitor	C20, C23	CAPC3216X180N	C1206C333J5GACTU	2
C0805C120J5GACTU	Capacitor	C26, C27	CAPC2012X88N	C0805C120J5GACTU	2
SS34-E3_57T	Schottky Diode	D1, D2	DIOM7959X262N	SS34-E3_57T	2
SMBJ15A	TVS Diode (Uni-directional)	D3	DIOM5436X244N	SMBJ15A	1
SMAJ5.0CA	TVS Diode (Bi-directional)	D4	DIONM5026X229N	SMAJ5.0CA	1
3403.0274.23	Fuse	F1, F2	3403027423	3403.0274.23	2
01000056Z	Undefined or Miscellaneous	F1B, F2B	01000056Z	01000056Z	2
BLM21PG221SN1D	Ferrite Bead	FB1	BEADC2012X105N	BLM21PG221SN1D	1
TB001-500-02BE	Connector	J1	TB001-500-02BE	TB001-500-02BE	1
SJ1-3515-SMT-TR	Connector	J3	SJ13515SMTTR	SJ1-3515-SMT-TR	1
1751248	Connector	J6	1751248	1751248	1
RLS-126	Power Supply	L1	RLS126	RLS-126	1
SRR4028-2R2Y	Inductor	L2	SRR4028	SRR4028-2R2Y	1
APT2012SGC	LED	LED1, LED2	LEDC2012X85N	APT2012SGC	2
IRM-20-12	Power Supply	PS1	IRM2012	IRM-20-12	1
RC0805FR-071KL	Resistor	R1, R8, R15, R17	RESC2012X60N	RC0805FR-071KL	4
ERJ-P08F1004V	Resistor	R2, R3, R4, R5	ERJP08_(1206)	ERJ-P08F1004V	4
ERA-6AEB103V	Resistor	R6	RESC2012X60N	ERA-6AEB103V	1
RC0805FR-0747KL	Resistor	R7	RESC1608X55N	RC0805FR-0747KL	1
ERA-6AEB203V	Resistor	R9, R12	RESC1608X55N	ERA-6AEB203V	2
ERA-6AEB133V	Resistor	R10, R14	RESC1608X55N	ERA-6AEB133V	2
ERA-6AHD100V	Resistor	R11, R13	RESC1608X55N	ERA-6AHD100V	2
RC0805FR-0710KL	Resistor	R16, R18	RESC1608X55N	RC0805FR-0710KL	2
B72660M0301K072	Varistor	RV1	CU4032	B72660M0301K072	1
5016	Test Point, 1-Pin SMT, RoHS, Tape and Reel	TP1, TP2, TP3, TP4, TP5, TP6	KEYSTONE-5016_V	CMP-2000-05178-1	6
R-78CK3.3-0.5	Regulador conmutado 3.3 V 0.5 A, SIP3	U1	R78CK3305	R-78CK3.3-0.5	1
AMC3330DWE	Integrated Circuit	U2	SOIC127P1030X265-16N	AMC3330DWE	1
ATM90E26-YU-R	Integrated Circuit	U3	SOP65P778X199-28N	ATM90E26-YU-R	1
ECS-81.92-12-30Q-EN-TR	Crystal or Oscillator	Y1	ECS81921230QENTR	ECS-81.92-12-30Q-EN-TR	1